CS190C Lec5

Build Transformer Decoder

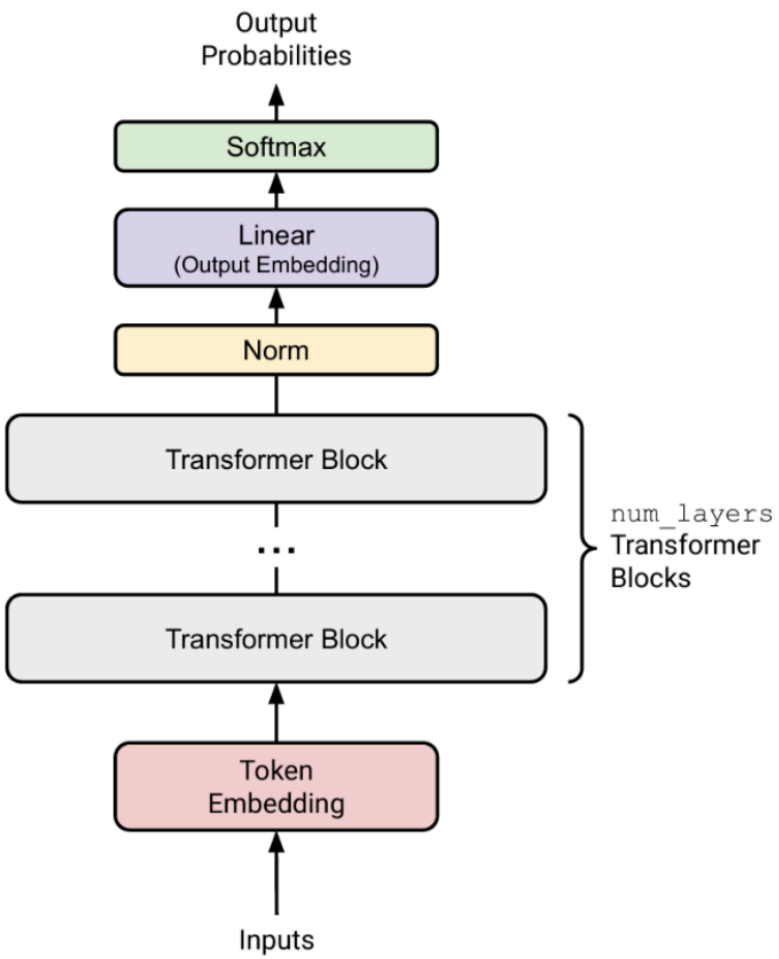
Attention please: The only content removed from these slides is the code implementation of each module.

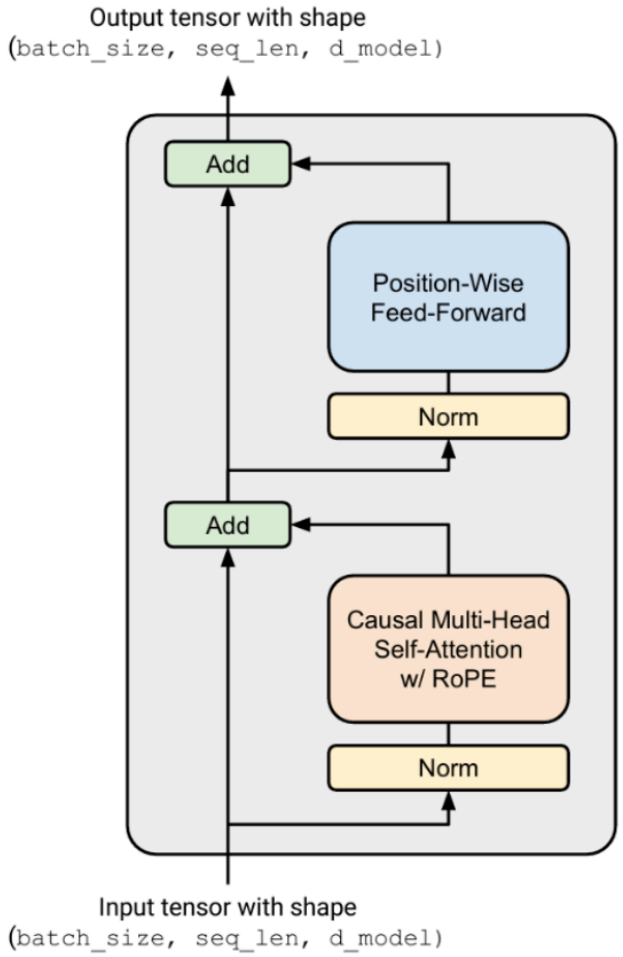
Overview

- Macro System Architecture
- Code Implementation of Basic Computational Components
- Code Implementation of RoPE
- Code Implementation of FFN and Attention
- Assembly of Transformer Block and Final Model

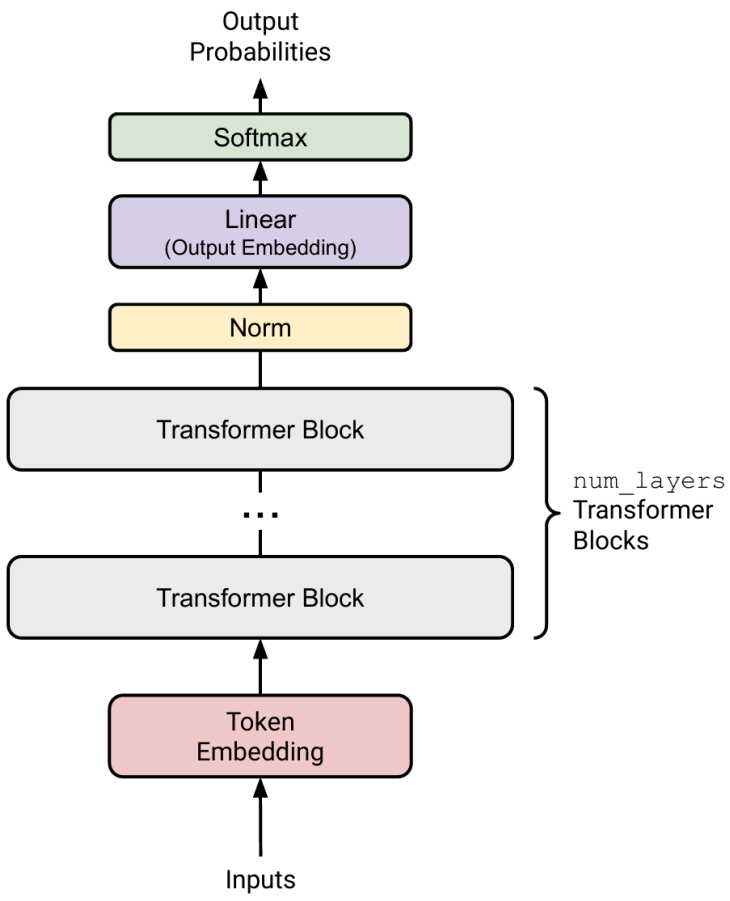
PART1: Macro System Architecture

Decoder-Only Transformer Overall Architecture Diagram



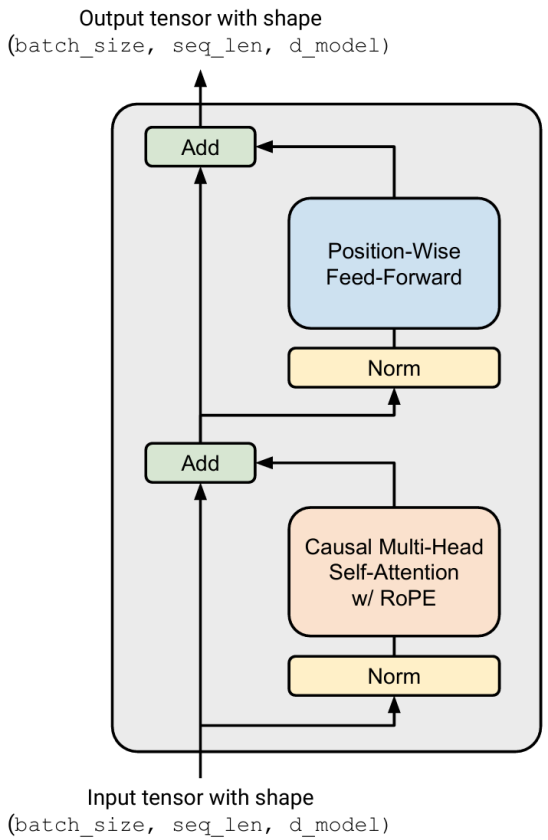


Modules of the whole Transformer



- Pass through a `Token Embedding` module to convert to a dense vector (Input layer)
- Pass through several `Transformer Block` modules to absorb information in multiple rounds (Hidden layer)
- Normalize the numerical scale of the final tensor
- Apply a linear transformation to calculate possible scores for generating each word (Output layer)

Modules of each Transformer Block

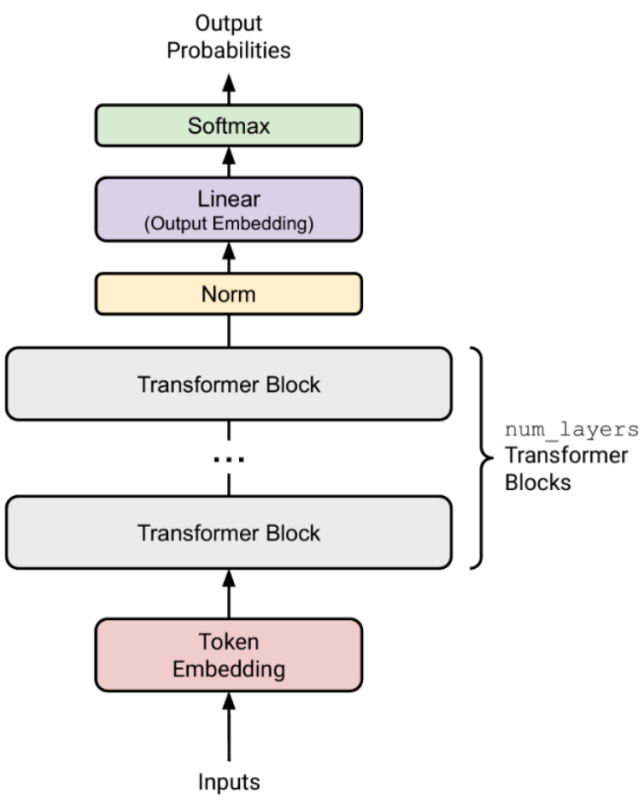


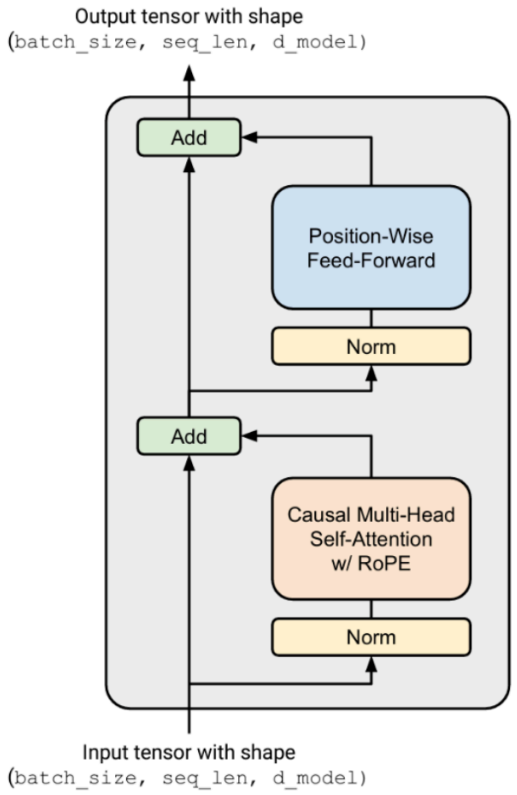
- Pre-layer RMSNorm module
- MHA Module with Residual Connection
  - contains a RoPE module if positional embedding is needed
  - contains a Softmax module for computing the attention score
- Pre-layer RMSNorm module
- FFN Module with Residual Connection
  - contains SiLU module

Moreover, almost all modules require using the **linear transformation** module!

PART2: Code Implementation of Basic Computational Components

Basic Components to be Implemented





Modules to be implemented:

- `Token Embedding`
- `Linear`
- `RMSNorm`
- `SiLU`
- `Softmax`

1. `class Generate_Embeddings`

IDEA:

- Most original input: BPE encoding results (e.g., `[3,10,2,6,4]`, all token IDs from the vocabulary)
- Tensor shape: `[batch_size, seq_len]`
- Expected model input: Different words in the vocabulary have different embedding vectors $\mathrm{emb}_i \in \mathbb{R}^{d_{\text{model}}}$
- Implementation idea:
  - Generate a learnable matrix $W_e \in \mathbb{R}^{|V| \times d_{\text{model}}}$.
  - The i-th column represents emb_i
  - Randomly initialized W_e, representing no prior knowledge about the meaning of any word at first.
  - Learn W_e with other parameters during the training process.

1. `class Generate_Embeddings`

Parameter scheme:

- Initialization phase: Pass in `vocab_size`, `d_model`, `device` (the device where PyTorch tensors are stored), and `dtype` (numerical type of tensor elements).
- Forward phase:
  - Pass in `token_ids` (shape `[batch_size, seq_len]`)
  - Output shape is `[batch_size, seq_len, d_model]`

1. `class Generate_Embeddings`

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2. `class Linear_Transform`

IDEA:

- Assume we need to transform a 3-dimensional tensor into a 6-dimensional one...
- Mathematically speaking: Let the 3D tensor `x` (1×3) right-multiply a 3×6 matrix `W`
- The shape of `xW` is then 1×6, just like the diagram below.

Question: Linear operations are the most frequent in LLMs...

Can this operation be accelerated as much as possible?

2. `class Linear_Transform`

- PyTorch tensors follow the "last dimension elements have contiguous memory addresses" principle.
- For a 3×6 tensor W, its memory layout is shown in the diagram (same color means contiguous):

- For a $4 \times 3 \times 6$ tensor (`[batch size, rows, columns]`), every 6 elements are also contiguous (e.g., the memory address difference between `[1,1,1]` and `[1,1,2]` is 1 unit, while the difference between `[1,1,1]` and `[1,2,1]` is 6 units).
- That is: Matrix elements in the same row have contiguous memory, while those in the same column do not $\Rightarrow$ **Row-major order principle**.

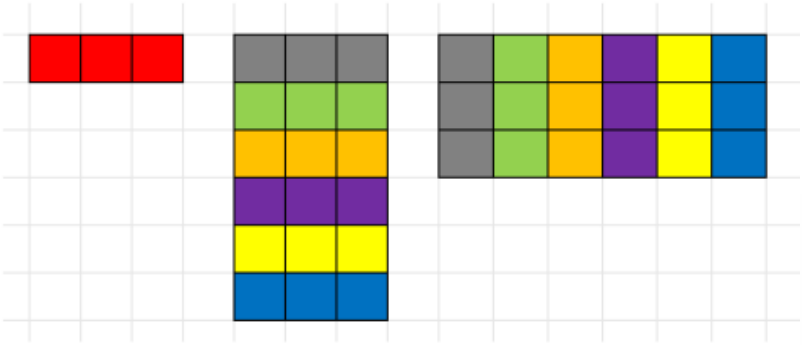
2. `class Linear_Transform`

Perform 6 vector dot product operations. In each operation:

- `x` participates as a whole row, memory is contiguous, can fully utilize cache
- `W` participates as a whole column, memory is not contiguous, may not utilize cache effectively

How can we make `W` also participate "as a whole row" in each operation?

2. `class Linear_Transform`



- PyTorch's transpose operation does not change the tensor's underlying memory layout, only the step size (Can you give an example?)
- Create new `W`: `[6,3]` (every 3 elements are contiguous in memory)
- Transpose `W` to `[3,6]`: Can perform matrix multiplication, and the memory distribution remains unchanged
- `x` and `W` have fully contiguous memory access during each operation, allowing full utilization of cache!

2. `class Linear_Transform`

```
[CODE EXPUNGED]
```

3. `class RMSNorm`

IDEA: Normalize the input tensor $\mathbf{a}$ (We've discussed why at `Lec3`)

- $\mathbf{a}_i = \frac{\mathbf{a}_i}{\mathrm{RMS}(\mathbf{a})} g_i$ (divide by normalization weight uniformly, and apply learnable fine-tuning)
- g_i is a learnable parameter
- $\mathrm{RMS}(\mathbf{a}) = \sqrt{(\frac{1}{d_{model}} \sum \mathbf{a}_i^2) + \epsilon} \quad \text{(Root Mean Square)}$

Input tensor shape: `[batch_size, seq_len, d_model]` $\Rightarrow$ Not a 1D vector, how to handle?

- **PyTorch's broadcasting mechanism**:
  - By default, operations are performed on the last few dimensions, while operations across earlier dimensions are equivalent to replication.
  - Equivalent to parallel operations on `batch_size * seq_len` parallel `d_model`-dimensional tensors.
  - Just process it as a 1D vector `[d_model]`!

3. `class RMSNorm`

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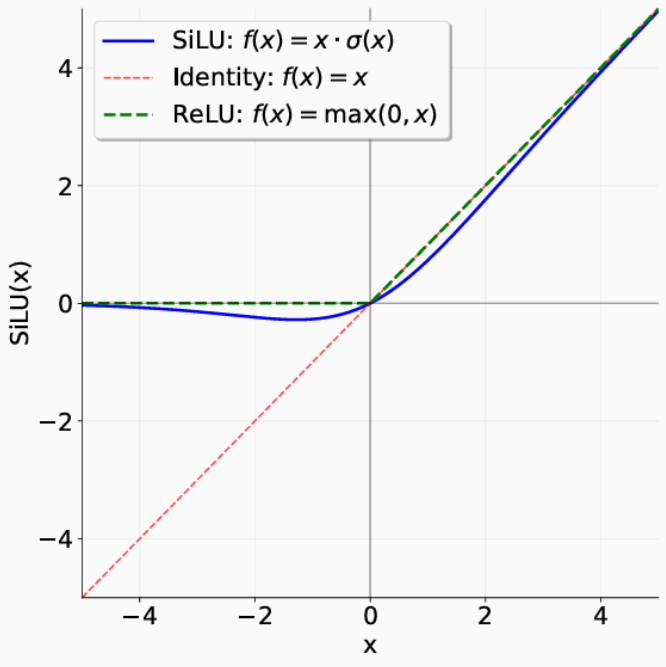
4. `class SiLU_Activation`



$$\text{SiLU}: f(x) = x \cdot \sigma(x)$$
$$\text{ReLU}: f(x) = \max(0, x)$$

- For $x < 0$: roughly equals to 0
- For $x > 0$: roughly stays the same
- Compare with ReLU: smooth and differentiable at $x = 0$

4. `class SiLU_Activation`

```
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5. `class Softmax_Activation`

How to turn a score tensor to a distribution?

$$x_i = \frac{e^{x_i}}{\sum e^{x_j}}$$

- Each x_i calculates its exponential as a weight, then performs weight normalization
- Can make the advantage of relatively larger values more pronounced
- Even smaller values remain non-zero after normalization

Problem: What if there exists a very large x_i? (e.g., normalizing `[20,3,1005]`)

- e^{1000} = `NAN`

5. `class Softmax_Activation`

- Normalizing `[100,101,102]` vs Normalizing `[-2,-1,0]`
- Weight of 102:

$$\frac{e^{102}}{e^{102}+e^{101}+e^{100}} = \frac{e^{0}}{e^{0}+e^{-1}+e^{-2}}$$

- The Softmax normalization result of `[100,101,102]` is equivalent to that of `[-2,-1,0]`

5. `class Softmax_Activation`

Let $x_{\max}$ be the maximum value among x_i:

$$\begin{aligned}\mathrm{Softmax}(x_i) &= \frac{e^{x_i}}{\sum e^{x_j}} \\ &= \frac{e^{x_i} / e^{x_{\max}}}{\sum e^{x_j} / e^{x_{\max}}} \\ &= \frac{e^{x_i - x_{\max}}}{\sum e^{x_j - x_{\max}}}\end{aligned}$$

Safe Softmax: Subtract $x_{\max}$ from all x_i to avoid problems with extremely large values that cannot be calculated!

5. `class Softmax_Activation`

```
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PART3: Code Implementation of RoPE

Review: RoPE Calculation Rules

Apply the rotation: $\boldsymbol{x} \mapsto \boldsymbol{R}^i \boldsymbol{x}$.

- Given an embedding $\boldsymbol{x} \in \mathbb{R}^d$ (where d is even) at position i:
- Divide the d-dimensional vector into $d/2$ pairs (2D sub-spaces).
- Rotate each pair by an angle $\theta_{i,k}$.

$$\boldsymbol{R}^i = \begin{bmatrix} \boldsymbol{R}_1^i & 0 & 0 & \cdots & 0 \\ 0 & \boldsymbol{R}_2^i & 0 & \cdots & 0 \\ 0 & 0 & \boldsymbol{R}_3^i & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \boldsymbol{R}_{d/2}^i \end{bmatrix}, \quad \text{where} \begin{cases} \boldsymbol{R}_k^i = \begin{bmatrix} \cos(\theta_{i,k}) & -\sin(\theta_{i,k}) \\ \sin(\theta_{i,k}) & \cos(\theta_{i,k}) \end{bmatrix}, \\ \theta_{i,k} = \dfrac{i}{\Theta^{2k/d}}. \end{cases}$$

Review: RoPE Calculation Rules

$$\boldsymbol{R}^i = \mathrm{diag}\left(\boldsymbol{R}_1^i, \boldsymbol{R}_2^i, \ldots, \boldsymbol{R}_{d/2}^i\right), \text{where } \boldsymbol{R}_k^i = \begin{bmatrix} \cos(\theta_{i,k}) & -\sin(\theta_{i,k}) \\ \sin(\theta_{i,k}) & \cos(\theta_{i,k}) \end{bmatrix},$$

The attention score with RoPE depends only on relative position.

- Property of rotation matrix $\boldsymbol{R}$: $\boldsymbol{R}^m(\boldsymbol{R}^n)^T = \boldsymbol{R}^{m-n}$
  - Transpose = inverse, Multiplication → Addition of powers.
- In attention mechanism, given a query $\boldsymbol{q}_i$ at position i and a key $\boldsymbol{k}_j$ at position j:

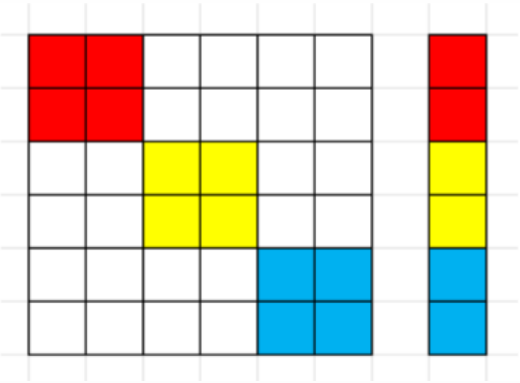
$$\boldsymbol{q}_i' = \boldsymbol{R}^i \boldsymbol{q}_i, \quad \boldsymbol{k}_j' = \boldsymbol{R}^j \boldsymbol{k}_j$$

$$(\boldsymbol{q}_i')^T \boldsymbol{k}_j' = \boldsymbol{q}_i^T (\boldsymbol{R}^i)^T \boldsymbol{R}^j \boldsymbol{k}_j = \boldsymbol{q}_i^T \boldsymbol{R}^{j-i} \boldsymbol{k}_j$$

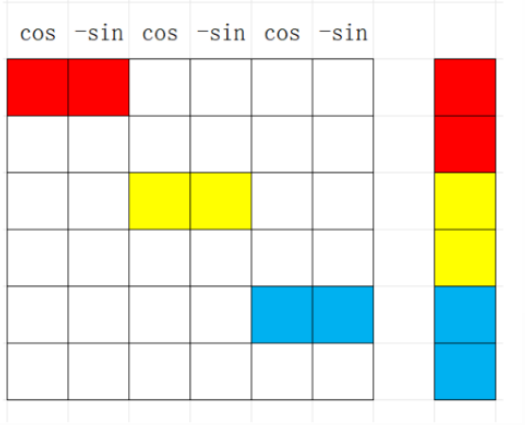
How to Avoid Brute-force Matrix Multiplication?

- Observation: Even rows of the matrix correspond to the same transformation rule: `[cos, -sin]`, with angle $\theta_{i,k}$ as the variable
- If these even rows that share the same rule can be computed efficiently in a unified manner, how should subsequent processing proceed?

$$R^i = \begin{bmatrix} R_1^i & 0 & 0 & \cdots & 0 \\ 0 & R_2^i & 0 & \cdots & 0 \\ 0 & 0 & R_3^i & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & R_{d/2}^i \end{bmatrix}$$

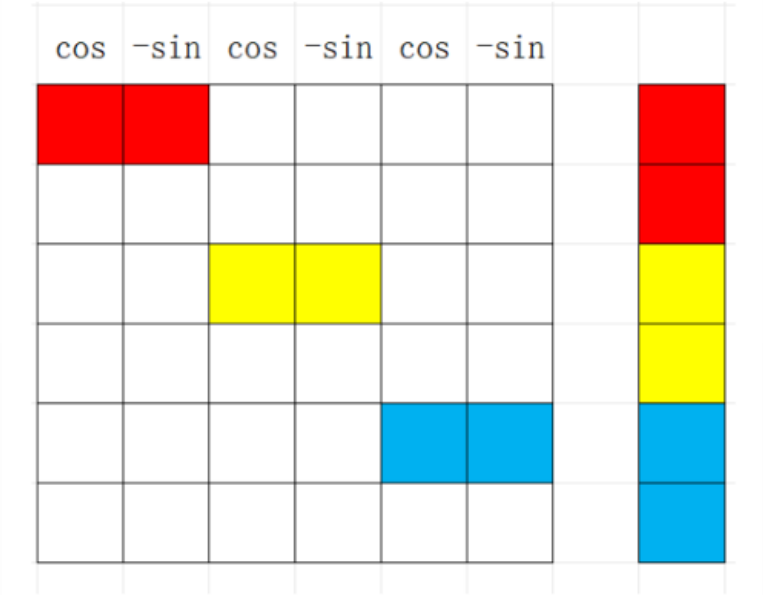


$$R_k^i = \begin{bmatrix} \cos(\theta_{i,k}) & -\sin(\theta_{i,k}) \\ \sin(\theta_{i,k}) & \cos(\theta_{i,k}) \end{bmatrix}.$$



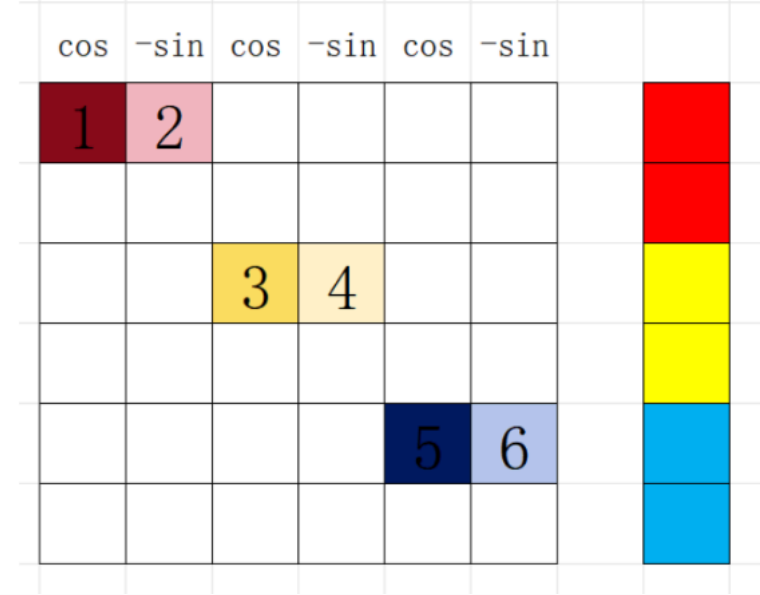
How to Avoid Brute-force Matrix Multiplication?

- Calculate products of `Red-Red`, `Yellow-Yellow`, `Blue-Blue`,
  - which correspond to: x_0, x_2, x_4 after RoPE
- The same applies to x_1, x_3, x_5.
- Construct 2-number blocks of $\boldsymbol{R}_i$, and calculate block-wise dot products.



How to Construct 2-number blocks of $\boldsymbol{R}_i$?

- Pre-calculate a $\theta_{i,k}$ lookup table of size `[max_seq_len, d/2]`
- Then calculate lookup tables of the same size for $\cos(\theta_{i,k})$ and $\sin(\theta_{i,k})$
- Obtain values at positions 1, 3, 5 directly from the cos lookup table
- Obtain values at positions 2, 4, 6 directly from the sin lookup table



How to Construct 2-number blocks of $\boldsymbol{R}_i$?

- Concatenate pairs `(1,2)`, `(3,4)`, `(5,6)` into 3 blocks, perform block-wise dot product with the 3 blocks of `x`, obtaining the values for the three even rows.
- Similarly, obtain the values for all odd rows of the transformed `x`, concatenate both to get the complete transformed `x` vector.

Code Implementation of RoPE

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```

Code Implementation of RoPE

```
[CODE EXPUNGED]
```

PART4: Code Implementation of FFN and Attention

1. `class Feed_Forward_Network`

Module algorithm (excluding residual connection):

- Input tensor `x` (`d_model` dimensional)
- Route 1: Transform via `W1` to `d_ff` dimensional, pass through `SiLU` activation function to get new `x`
- Route 2: Transform via `W3` for another expansion, obtaining another `d_ff`-dimensional tensor
- Perform element-wise multiplication of two tensors to obtain a new `x`
- Transform back to `d_model` dimensional via `W2`

We call it `SwiGLU`:

$$\mathrm{FFN}(x) = \mathrm{SwiGLU}(x, W_1, W_2, W_3)$$

1. `class Feed_Forward_Network`

```
[CODE EXPUNGED]
```

1. `class Feed_Forward_Network`

```
[CODE EXPUNGED]
```

2. `class Multihead_Attention`

Module algorithm (excluding residual connection):

- Input tensor `x` with size `[batch_size, seq_len, d_model]`
- Through three linear transformations, each `d_model`-dimensional word vector of `x` is projected to obtain:
  - `Q`, `K` matrices `[batch_size, seq_len, num_heads*d_k]`
  - `V` matrix `[batch_size, seq_len, num_heads*d_v]`
- Perform `RoPE` positional encoding on `Q`, `K` matrices
- Generate attention upper triangular mask (to prevent breaking the "autoregressive" assumption)
- Use QKV matrices and attention mask to compute attention output

2. `class Multihead_Attention`

Method for calculating attention output:

- Multiply `Q`, `K` matrices to calculate token feature matching scores
- Apply mask processing to the matching score matrix
- $\mathrm{Softmax}(\frac{QK^T}{\sqrt{d_k}})$, calculate attention scores between tokens
- Multiply the score matrix by the `V` matrix to get the attention output of each head
- Multiply by the `W_O` matrix to integrate the heads

Calculation of Attention Output for One Head

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Generation of Attention Mask

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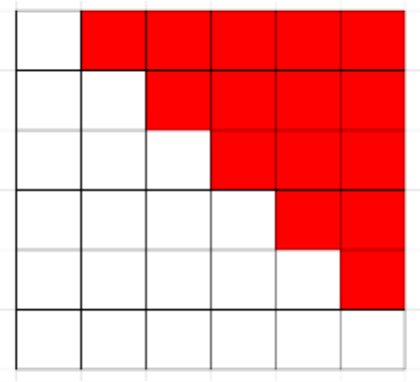

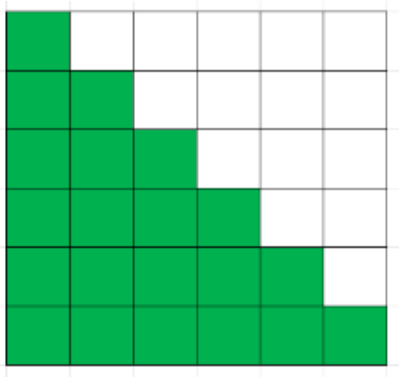

List of Sub-modules in Multihead_Attention

- Attention mask generation module
- Attention output calculation module (SDPA)
- RoPE module $\Rightarrow$ Requires additional parameters like `max_seq_len`, `theta`, `token_positions`, etc.
- Four types of linear transformations: `Q`, `K`, `V`, `O`

Assembly of Complete Module

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Assembly of Complete Module

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Assembly of Complete Module

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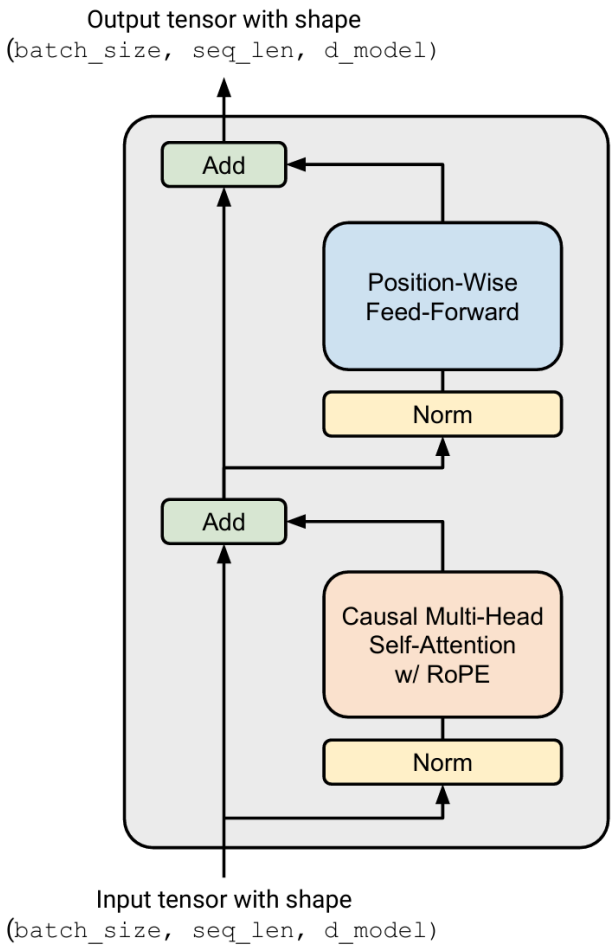
PART5: Assembly of Transformer Block and Final Model

Structure of Transformer Block



- An RMSNorm module
- An MHA module
- Residual connection
- Another RMSNorm module
- An FFN module
- Residual connection

The parameters received by each Block consist of all parameters required by the modules listed above!

Assembly of Transformer Block

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Assembly of Transformer Block

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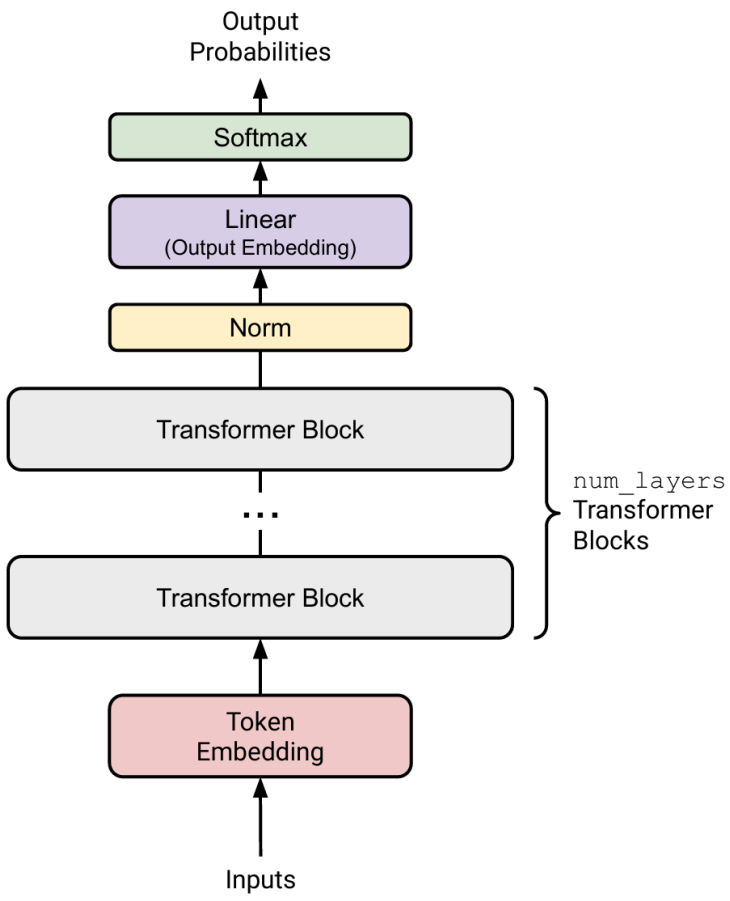
Structure of Full Transformer



- A Token Embedding module
- Several Transformer Block modules
- A RMSNorm module
- The Final Linear Transformation module

Assembly of Complete Transformer

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Assembly of Complete Transformer

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